
Innovation in Grand Prix motor racing: the evolution of technology, organization and strategy

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Abstract

Since its inception at the end of the 19th century, motor racing has been linked with the automobile industry. Grand Prix racing has contributed not only to the transfer of technology into roadcar manufacture but also to the advertising and promotion of cars and components. Until the mid-1950s, the relationship was dominated by motor manufacturers, who used race tracks as an extension of their R&D facilities. About 1960 the relationship became more complex, characterized by a network of constructor organizations and sponsors as well as manufacturers. Entrepreneurship has become diffused within this network which pursues several objectives: primarily winning races and transferring promotional benefits to sponsors. The use of the track as a test bed for technical innovation has remained important but is secondary. Organizational and commercial, as well as technological, innovations have led to the development of both motor sport and a Grand Prix industry.

1. Overview

Motor racing at its simplest is no more than a collection of vehicles competing over a measured distance. For the Grand Prix racing enthusiast, however, it is intimately and intricately involved with the implications of different driver/car combinations, practice ses-

sions, lap times and grid positions. The interest of the technical specialist extends further still, to embrace the comparison of engine configurations, chassis and suspension layouts and aerodynamic advantages. However, while these factors provide the basis of its sporting interest and commercial success, Grand Prix motor racing encompasses wider considerations yet.

The history of this category of motor racing is linked to the development of the automobile itself and the industries which stem from it, the flow of technological innovations throughout those and other industries, global economic fluctuations and the vagaries of war that have both helped and hindered these developments, and the emergence and growth of a motor racing 'industry' based firmly upon sustained innovation—in organization and marketing as well as in technology.

Our study examines the changing relationships between motor racing and production car manufacturers as Grand Prix racing has developed from a sport into an industry. In particular, we have been concerned with the changing status of motor racing from a source of technological innovation for car manufacturers to the focus of a complex industry in its own right based on innovations in organization and strategy. A full report of the project is available from the first author. Entitled *Technology, Marketing and Organization in Grand Prix Motor Racing Innovation: Strategic History and Case Analysis*, it describes the methodology employed, case studies of some recent constructor organizations and a more detailed account of the strategic implications of the development of Grand Prix motor racing into a 20th century industry. The present paper provides the historical background to the case studies but stands alone as an account of the evolution not only of the technologies on which the transfer of core innovations was dependent but of the organizational and strategic developments which relied equally upon the availability and transfer of complementary assets relating to marketing and promotion.

2. The rise of the manufacturers

Motor sport can be traced back to the first days of the automobile: as early as 1878 a race for 'Steamers' was held in the United States. In

1897, 2 years after Benz's production of the first petrol-engined vehicle, an unsuccessful sprint took place in France, though it was not until 1895 that motor racing became established in Europe as a sport. The first recorded Grand Prix, a 2-day event run on a closed circuit near Le Mans, was staged in June 1906. (It was officially designated the Ninth French Grand Prix by the Automobile Club of France, A.C.F., in deference to eight previous inner-city races, run on public roads and marked by several fatalities, that had been organized by that body.) That 'first' Grand Prix was won by a Renault. Paradoxically, the French car was judged technically inferior to previously raced models: whereas, prior to 1906, Renault had relied on a chassis that consisted of stressed skins riveted to the base, the first semi-'monocoque', or 'single shell' which became prominent in later years, the Grand Prix car had a more orthodox chassis. The victory is generally attributed to the skill of the driver rather than any innovative design or device. The reputation of the Grands Prix's status was soon to be re-established on technological advance rather than on novelty and spectacle, as the manufacturers comprising the rapidly developing automobile industry came to dominate the sport [1; for the background to the national motor industries concerned, see *inter alia* 2–13].

Indeed, the inaugural years of Grand Prix racing came to be dominated by the infant European manufacturers. Among others, Renault, Fiat, Panhard, Mercedes Benz and Peugeot participated in the earliest races and dictated the pace of technological progress [14, 15, p. 8]. The machines they raced were from the outset virtually identical to the road-going 'production' models and were similarly based on the technology of wooden coach-building design and construction. However, as motor sport increased in public popularity manufacturers built cars specifically for competitive racing: success in competition led to increases in car sales and a manufacturer's status was

linked directly to victory on the track. As more manufacturers entered the industry both the number of races and the number of vehicles per race increased proportionally; the search for improvement in performance became paramount [16,17, pp. 117–20]. By the early years of the century, motor racing was playing its part in overcoming remaining prejudice against the internal combustion engine.

Crude as these machines now seem, many were endowed with refinements which indicate the rate of technical progress at the time—their good handling, stability and cornering power were lacked by roadgoing models—but most suffered still from under-development: in tyres (where the technology was unable to keep pace with the increased loads they bore), materials (whose properties were as yet unknown and which were poor in quality) and lubrication systems (which, at best, were primitive) [14, p. 73]. However, one technical consideration, engine size, was given pre-eminence in the quest for speed and victory. The first, traditionally-engineered Grand Prix cars varied enormously in the size of their engines which ranged from 8 to 18 litres displacement. To accommodate them, chassis were constructed with:

“such generosity of metal as was judged appropriate to races that might endure for 2 days over public roads that were poor to begin with and would deteriorate rapidly as the mighty cars passed” [14, p. 20]

Regulations have played a critical part in determining the technological progress of the cars participating in Grand Prix racing ever since its inception. The technical characteristics of cars belonging to the various classes of motor sport (the Formula One of Grand Prix racing, and the other, less prestigious formulae) are denominated by such official regulations. Moreover, the quest for speed and, later, commercial returns has always had to be conducted within the limitations of rules imposed by the sport's governing bodies. An

early regulatory intervention with notable technical and commercial ramifications took place in 1908 when the A.C.F. stipulated that Grand Prix engines should have a maximum piston area and the vehicle a maximum overall weight: the technical impact was to reduce the speeds of which the huge machines then produced were capable; the resulting commercial impetus was towards ingenious alternative means of delivering the power required to win. Moreover, the limitation of piston capacity prompted manufacturers to examine critically the emerging technical concepts and to reassess the implications of compression ratios, combustion efficiency and piston design. The imposition of a weight limit also inspired investigation of chassis development and weight distribution to optimize efficiency [1, p. 38;14, p. 24].

The Grands Prix of the following years, now generally known as the ‘Grands Prix de l’A.C.F.’, grew in prestige as the most significant motor sport events of the period. National pride, as well as industrial prestige, was linked to the achievement of victory on the track. However, Renault's losing in two consecutive years' races, coupled with the escalating costs of building these highly specialized vehicles and a general trade recession that exacted a toll on car sales, led the A.C.F. to allow the Grand Prix to lapse [1, pp. 38–45]. Other forms of racing continued, nevertheless, to thrive, notably the ‘voiturette’ class, and when the Grands Prix were re-established in 1912, a Peugeot voiturette beat the larger cars [15, p. 49]. The victory spurred the development of technology as a source of competitive advantage and by 1914 most new engines were based on Henry's inspired design of twin overhead camshafts and four valves per cylinder engine [14, p. 34].

World War I halted all motor racing, but contributed, nonetheless, to technical developments which in the longer term were responsible for the enhancement of the sport. These included progress in materials, fuel and

carburation, cooling and component design as well as advances in supercharging and turbo-charging derived from the emerging aircraft industry. The pattern of adoption of the new technology was precisely that recorded repeatedly in the study of the diffusion of innovations: some automobile designers, including Bugatti and Bentley, took up the new technologies immediately but "in general, the motor racing elite affected a disdainful ignorance of high performance aero technology" [14, p. 55].

Bugatti's design for aero engines, though not successful, was adopted and adapted by engine designers in the United States where the 'straight eight' layout (which could be increased in multiples of two cylinders to increase size) was in postwar use. Partly as a result of having suffered only one year of warfare, the United States became the dominant influence at this time. When the Grands Prix de l'A.C.F. were reintroduced in 1921, the race was based on the American-inspired straight eight 3-litre engine limit. The technological advances achieved during the war years are encapsulated in the outcome of this race: the Henry-designed engine, run in a Ballot car, was unreliable at high speeds due to poor bearing surfaces and lubrication, whereas the winning Duesenberg benefited from a refined lubrication system, greater capacity to produce power and higher piston speeds which significantly enhanced engine responsiveness. The Duesenberg also had the advantage of an efficient hydraulic braking system which used water as the working fluid [14, p. 73]. The straight-eight concept was developed to greater efficiency by Fiat who designed an engine with both main bearings and big-end bearings of the roller type, and used two valves per cylinder instead of four, allowing a higher revving and more compact engine.

Throughout the increasing popularity of motor sport events and the creation of purpose-built tracks on which to stage them, the involvement of manufacturers remained

prominent. They contributed throughout this period to the advertising and promotion of motor racing, technical development and, not least, national pride. Engine and chassis technologies continued to make progress but other areas were now being seriously investigated in terms of their contribution to competitive advantage. In 1922, a change in the regulations by the successor to the A.C.F., the Commission Sportive Internationale (C.S.I.) banned the carrying of a spare wheel which prompted an investigation of tyre wear rates and compounds that could endure the whole race. There was a growing awareness that a reduction in drag (air resistance) could increase speed and the Grand Prix of that year saw a number of entrants sporting streamlined bodywork: Bugatti, for instance, had fitted a rounded front and a long tubular tail. Aerodynamics had been considered as early as 1914 and received some impetus as a result of the development work that took place during the war, but advances based on the new technology were not taken up in earnest until the 1920s [14].

Grand Prix technology had progressed rapidly: in the space of 17 years it had outpaced the original cars in both speed and technical development. The Fiat Type 805, which incorporated numerous technical advancements based on the accelerating development of automotive technology, and which in 1923 reintroduced the supercharged engine to Grand Prix competition [15, p. 50]:

"left almost the entire racing world anxious to endorse Fiat's engineering policy. Not only had the principles of engine design been radically revised, but also those of body shaping and even, to some extent, chassis design" [14, p. 78].

Engine development, nevertheless, took precedence over other areas of potential advance [14, p. 100], though up to the outbreak of World War II, technical refinement continued in such areas as carburation,

supercharging, aerodynamics and suspension layouts.

The patriotic significance of Grand Prix success, manifest as early as 1907 when competing cars were required to wear the allotted racing colours of their respective countries of origin, gave way during the 1920s and 1930s to a more serious nationalism which, in turn, affected the course of technical progress. In the early 1920s, the Italian motor industry, under Mussolini's influence, concentrated its efforts on the Alfa Romeo Grand Prix cars and was successful in using the competition in the interests of national prestige. This idea was later taken up by Germany which pursued an explicit policy of promoting National Socialist ideals through technological accomplishments. Subsidies were paid to Mercedes Benz and Auto Union (the two main national Grand Prix teams) for development work which could be offset against the rearmament budget, and a prize was offered for the most successful German Grand Prix car of 1934 [18, pp. 26–30]. These forces undoubtedly promoted the development and understanding of the Grand Prix car to the extent that both German teams dominated the latter part of the 1930s. A revised formula, the 750 kg formula, produced cars that were amongst the most powerful ever built and arguably as fast as modern vehicles [19, 27]. The political encouragement to further technological improvements undoubtedly produced more sophisticated designs, the use of lighter materials and a significant input from aircraft technology, now firmly established, in the form of greater aerodynamic considerations [20, p. 237].

Grand Prix racing emerged from World War II in a state of predictable uncertainty. The international body that governed all motor sport, once the Association Internationale des Automobile Clubs Reconnus, now known as the Fédération Internationale de l'Automobile (F.I.A.) was reformed and promptly agreed a new set of regulations, allowing 1.5-litre supercharged and 4.5-litre unsupercharged engines

[18, pp. 112–14]. Improvements in engine output can be gauged by the following comparison: the Alfa Romeo supercharged type 158 in 1938 produced some 190 b.h.p. at 6500 r.p.m. A year later, output reached 225 b.h.p. and by 1948 over 310 b.h.p. could be achieved. By this time, technical innovation had reached high levels of sophistication, much of the impetus coming from the war effort: for example, the large 4.5-litre Ferrari engine, through careful design to eliminate stresses, was lighter than the compact 1.5-litre supercharged engine in the British B.R.M. The Alfa Romeo type 159 supercharged engine could produce 385 b.h.p. at 9500 r.p.m. with three times the normal atmospheric pressure and 404 b.h.p. if pushed to 10 500 r.p.m. [20, p. 222].

As if to underline the critical role of the major manufacturers since the inception of Grand Prix racing, the sport's competitive nature was to be its undoing at this time. With a great sense of national pride still apparent in the sport, as well as the commercial criteria, the domination of the Italian Ferrari team over fellow Italian Alfa Romeo prompted the Alfa concern to withdraw from the scene, recognizing that its supercharged engine offered no further scope for development and deciding to rest on its previous achievements [18, p. 202]. The British B.R.M. team was struggling with an uncompetitive engine, other teams were similarly uncompetitive and prospective entrants were unforthcoming, when the F.I.A. announced a new set of regulations to come into effect 2 years thence. This measure reduced the probability of new entrants' coming forward immediately since there was no point in incurring the expense of developing a car with a lifetime of only two years.

The Grand Prix emerged again in 1954, governed by the new 2.5-litre regulations. The new cars fielded—Maseratis, Lancias, Coopers, Vanwalls, Connaughts and H.W.M.s—had been racing in the old voiturette class, now the less expensive Formula Two class. Because they used less powerful engines,

their entry to the Grands Prix engendered another search for enhanced performance based this time on improvements to chassis, suspension and running gear. The resulting innovations were many and varied: for example, the Lancia D50 used a V8 engine as an integral part of the chassis instead of its being fitted inside it; and Mercedes Benz produced the W196 which used an aircraft-based fuel injection system for the first time in a Grand Prix car, new swing-axle rear suspension and in-board drum brakes front and rear as well as an enclosed, streamlined body [21, p. 32; 22, pp. 569–599; 23, pp. 118–126; 24].

3. The coming of the constructors

The mid to late 1950s witnessed the emergence of several forces that were greatly to influence the future of the formula: first, the increased technical efforts, already described, that rapidly changed car performance; secondly, the upsurge of interest in the sport, particularly in Britain where new cars and drivers were challenging the established winners; and, finally, the beginnings of a greater commercial influence that was, within a few years, to pervade Grand Prix racing.

By the mid 1950s, motor racing commanded great public interest, due in no small part to the emergence of new cars and drivers, but also to the surprise win by a British Connaught, with a British driver, in the 1955 Syracuse Grand Prix. The next several seasons saw an upsurge in public awareness of the British racing effort as well as the more prominent position of commercial advertising. It had always been understood within the sport that commercial enterprise played a major part in its evolution and the early manufacturers were quick to seize on the sport's image to promote their products. Given their aim of sustained success on the track, these manufacturers had been forced to pursue technological innovation as well as commercial effectiveness. In the late 1950s, the climate changed: Grand Prix racing was an

established spectator sport with its own international governing organization [18, p. 244]. British fortunes excelled as European interest declined (due to a spectacular disaster at Le Mans in 1955) and in this situation the characteristics of the sport were fundamentally changed in two ways. The first was the fact that:

“the centre of the motor racing world had shifted from Italy to Britain, where specialist chassis-building teams used off-the-shelf engines and transmissions supplied by specialists. Only Ferrari and B.R.M. of the regular runners were left building their own chassis, engines and gearboxes, and the new era of the Formula One ‘special builders’ had arrived” [14, p. 242; 25, pp. 204–205].

The second was an impending change in regulations, strenuously resisted by the racing organizations, drivers and manufacturers, but successfully championed and encouraged by the participating oil companies which wished to increase public awareness of their products through their connections with motor racing. Prior to this time, fuel was mixed with nitromethanol but, as from 1958, only petrol was allowed, a significant milestone in the direction Grand Prix racing was about to take.

The characteristics of the sport were now such that “the racing was done by little firms making little cars, the simplicity of their design and crudeness of their construction bearing witness to the paucity of their resources. Very few cars were the work of one man or even a single team of men unified by the loyalties or enthusiasms of a common employer; discrete bands of specialists from various companies within the motor industry now supplied special needs with highly specialised products, so that the design of a racing car might be the joint work of engineers in many different firms, each specialising in certain components—brakes, tyres, gearboxes, engines, ignition systems, dampers and the like . . .” [25, p. 243]. In

addition, the cars themselves took on a revolutionary redesign as a result of the amalgamation of a small Formula Three car (a Cooper) and a lightweight engine (manufactured by Coventry Climax) that had once been the basis of a fire pump. The Formula Three Coopers were very light, handled well, had a low frontal area which improved speed and, most importantly, had the engine fitted in the rear. This idea was not new—Auto Union had tried it earlier—but in the Cooper it allowed a small cross-sectional body determined only by the dimensions of the engine and allowed the total body area to be reduced to a minimum, thereby reducing the drag on the car. The Cooper also used coil springs instead of leaf springs and had adjustable suspension.

Just as Cooper was transforming the layout of the Grand Prix car, the commercial revolution was beginning to gain momentum within the sport: a trend that had begun with the oil companies was spreading so that those who underwrote the sport financially were buying the right to dictate the terms on which it operated. Teams and drivers were courted now by the interested companies and became contracted to them. Grand Prix racing began to see the stirrings of a commercialism that was to have a profound effect on its development during and well beyond the next decade. The revolutionary Cooper design set precedents for Grand Prix car layout: the advantageous location of the engine at the rear, the incorporation of sophisticated and adjustable independent suspensions, chassis which, though still tubular in construction, were lighter, stronger and more efficient. A change in regulations in 1961 also stipulated other measures, mostly on safety grounds, such as dual braking systems, a 'roll over' steel hoop for driver protection, easy cockpit access, and the fitting of catch tanks to engines to prevent oil spillage. With the implementation of regulations governing the type of fuel permitted, all teams were forced to modify their engines and with the introduction of a new formula—limiting engine capacity to a

maximum of 1.5 litres—many teams ceased manufacture of their own engines and used the commercially available Coventry Climax unit. As a result they became dependent upon that company's continued support. For those teams whose resources allowed independent development—Ferrari, Porsche, and the sole British manufacturer, B.R.M.—there were no such problems.

The newer types of organization that had emerged as Grand Prix teams—Lotus, Brabham and Lola—relied on external manufacturers for parts, including engines, and on sponsors, from within and beyond the automotive industry, for financial support. They were destined to show the way forward for Grand Prix racing in general: initiated by men with a passion for the sport, organizations like Lotus were to bring the sport to maturity through their economically determined commercial links with external sponsors. This may explain why the 1.5 litre formula and subsequent formulae prompted innovative changes. The great similarity of the car designs of highly competitive teams in the early 1960s attests to the fact that those teams were unlikely to be able to secure further comparative advantage by greater fundamental exploitation of the general principles of automotive engineering. The advantages were sought rather in the comparatively continuous development of components: tyres, electronic ignition systems, fuel injection, improved spark plug technology, monocoque chassis design and aerodynamic efficiency. As a result, while innovation still occurred, and there was continual debate over the minutiae of technical aspects, the sport began to settle into a phase of incremental technical development.

Just as these technical advances led to the refinement of the Grand Prix car, other developments were to end in blind alleys. Among these were front-wheel drive, four-wheel drive, six-wheeled cars and gas-turbine propelled cars, all of which were found, usually after a great deal of development work, to

perform no better than the conventional alternative. Organizations in pursuit of success had to balance the possible advantages of radical changes, and the risks of failure inherent within them, against the need for the more predictable effectiveness on the track that would attract sponsors.

Several vital changes took place in the mid 1960s. The 1.5-litre car, though highly developed, was considered easy to drive and F.I.A. introduced a new regulation to the effect that from January 1966 the maximum engine capacity would be 3 litres. Advances in engine technology led to the prediction that the new capacity would lead to very high power output, up to 500 b.h.p. after some development. This figure was based on existing technology being able to produce 210 b.h.p. at 10500 r.p.m. from the 1.5-litre Coventry Climax engine and though tyre development saw improved grip and traction some engineers predicted the necessity for the four-wheel drive to cope with the power.

Coventry Climax announced in 1965 its withdrawal from Grand Prix racing as a result of the prohibitive escalation of costs; and a 'tyre war' developed among the manufacturers Firestone, Goodyear and Dunlop which effected rapid change in tyre compounds, profiles and construction, to the extent that for a time improvements in tyre efficiency outpaced those in car performance. This commercial battle resulted in tyres of unusual dimension and contours as each manufacturer attempted to gain greater track adhesion and transmit more power from the rear wheels to the road.

As such explicitly commercial elements came to pervade Grand Prix racing, so more constructor organizations entered the industry. In 1964, the Japanese motor cycle manufacturer, Honda, entered the sport and contested four seasons with moderate success. In addition, the American-based Eagle car, the construction of which was unique in the extent to which it incorporated light metal alloys, became another new entrant. However, the development which

would have the most far-reaching ramifications occurred in 1967 with the introduction of the Cosworth D.F.V. V8 engine. This engine is significant not only for its technical and design concepts but in that it was conceived and produced as a result of the coming together of the technical vision of an 'enthusiast', Colin Chapman, and the commercial criteria determined by a manufacturer, Ford of Britain, in deciding to promote its 'Total Performance Concept' image via this involvement in motor sport [17, p. 119]. The immediate success of the Lotus/Ford/Cosworth partnership had been immaculately planned, with close liaison between Lotus and Cosworth over the chassis/engine design and dimensions. In the course of more than two decades, the D.F.V. exerted a determinative influence on the progression of Grand Prix racing. Not least in technological terms, its constant development and use in other formulae showed its design concept to be remarkably robust, having given rise in the course of over two decades to several derivative engine designs [26, pp. 198–203; 27, pp. 215–217].

It is at this period that Grand Prix racing can be said to have assimilated a commercial rationale, to have become accepted in the sport, and to have exploited itself fully. Gone were the defined colours of the participating nations, replaced by promotional decals carrying sponsors' names; the cars now wore associated colour schemes. Another landmark was the recognition by commercial organizations that at the international level the spectacle of a Grand Prix was of enormous public interest which led to the further awareness that the sport could be used for promotional and advertising purposes. As if to reinforce these new forms of interactive innovation, Chapman concluded a sponsorship deal with John Player and Son which went beyond the previously accepted boundaries of sponsorship in that it was not confined within the automotive industry; it was also on a larger scale than any preceding sponsorship arrangement. The banning of

tobacco advertising on television in the 1960s had prompted John Player to seek a new promotional medium. With the appearance of the Lotus 49 in 1968, sporting the 'Gold Leaf' name and colours, Chapman and John Player emulated the involvement of Ford of Britain in motor sport: the enhancement of commercial objectives by association with Formula One racing. The significance of the move was that John Player were in an entirely separate industry and the commercial implications of the new relationship between a constructor and a sponsor were not lost on other teams and companies outside the motor trade. As more Grand Prix teams emerged, as television extended its coverage of motor sport events and as the medium of sponsorship was appreciated more widely as a cost-effective way of promoting a corporate image, so the Grand Prix organizations, their structure, character and nature, evolved into a motor racing *industry*.

In a move which was to have ramifications throughout Grand Prix motor sport for two decades, the Ford Cosworth V8 engine was made available in 1968, at a commercially viable price, to all participating Grand Prix teams. This strategy had three major implications for the sport. First, since adopting teams were now using engines comparable in power output and performance potential, developmental work turned towards the engineering of greater efficiency in power transmission and hence concentrated on the design and operation of the chassis. Secondly, the sport was further opened up through the availability of so major a component as the engine that could be purchased 'off-the-shelf' from an external manufacturer; some Grand Prix cars were dubbed 'kit cars' for this very reason. Thirdly, with this element of commonality among the teams, the need for constant development, to achieve even a small competitive gain, became urgent; most teams therefore showed increasing awareness of the need to balance experimentation and innovation with the practicalities of week-to-week rac-

ing schedules. The technical development of the cars came from various quarters and, almost without exception, individual teams were no longer responsible for the total design and manufacture of a Grand Prix car. A growing number of peripheral agencies assisted with technical expertise with regard to the performance of their own product within the car [25, p. 217; 28 *passim*].

While the Ford Cosworth V8 may have established itself as the most reliable and, ultimately, the most successful on the track, other makes of engine contested the 3-litre formula: they were all V configurations with either eight or 12 cylinders and various levels of power output, though reliability was the most important criterion. Newcomers to the Grand Prix scene were the British McLaren (another instance of a driver turned constructor), the French Matra concern and the British March team. Numerous detail changes are recorded for this period of the late 1960s–early 1970s as each team sought to improve its relative performance, but the most significant innovation was in the use of aerofoils.

Aerofoils or 'wings' first appeared on a Grand Prix car in 1968 in order to increase the downward thrust on the tyres and achieve greater grip. The concept, which is the basis of aeroplane flight, had been applied to some earlier racing cars, notably a Porsche in 1955 and a Chaparral in 1968; however, the idea was now taken up in earnest by Grand Prix teams. Tyre development had continued to make significant improvements in grip, and the introduction of aerofoils allowed the enhanced tyres to work at maximum efficiency by exploiting their traction and cornering capabilities. However, while aerofoils improved speed through corners they also reduced straight-line speed by increasing the aerodynamic drag which is an inescapable concomitant of aerodynamic lift [29]. Efforts were made to balance these forces in order to attain optimal overall speed, and with this technical development, the engine, chassis,

suspension, tyres and aerofoils had all to be finely tuned to perform in complete harmony. The result was that the Grand Prix car reached greater levels of sophistication.

During the following 10 years a large volume of research and development activity was directed towards the further harnessing of aerodynamic effects to the benefit of Grand Prix racing, culminating in the advent of the ultra-sophisticated 'ground-effects' cars. Wing technology was at first crude. The wings were light and poorly mounted so that they often broke with disastrous consequences. They were not always mounted in the most aerodynamically efficient position, sometimes acting against the chassis rather than with it. They were fixed devices with no facility for any form of tuning and often stood high above the car to gain maximum downforce. However, within a matter of months more flexible versions, internally adjustable and of differing shape and cross-section, were devised and incorporated. Eventually, though only after regulations governing the formula had limited their dimensions and locations in the interests of safety, wings became an integral part of the Grand Prix car's bodywork.

The early 1970s, the period of the kit car constructors, also saw considerable theoretical developments in the fields of chassis design and aerodynamics. The new design philosophies resulted in wedge-shaped bodywork on many cars such as the Lotus 72 and the Ferrari 312B. With engines now producing power output of typically 450 b.h.p., the emphasis continued to be on chassis efficiency and improved grip: many teams effected detail work to overcome the increasing problems of the opposing forces that plagued cars, like aerodynamic lift and downforce. Bodywork began to be all-enveloping with large noses appearing on the front of many cars to accommodate cooling radiators as well as acting as an aerofoil. While this avenue seemed to hold most promise for constructors, it also had its drawbacks: streamlined bodywork could induce an

undesirable amount of 'lift' (upforce) which could affect the stability of the car; air flow was needed to cool tyres, brakes and engine/oil radiators, which if enclosed by bodywork required several built-in ducting systems; in the event of repairs being needed the bodywork prevented accessibility to such parts and wasted time. Therefore any advantages gained in aerodynamic styling could easily be negated by other considerations.

The similarities in design between cars of this period bear witness to the tendency towards uniform technical thinking of the time, though it must be remembered that design is a response to the prevailing regulations, many of which act against further development; the requirement that all Grand Prix cars have uncovered wheels, for instance, eliminates fully aerodynamic body styling. However, commercial and competitive pressures also contributed to conformity as the teams and companies sought to exploit the same technological opportunities. The attempt to increase the traction and downforce on car tyres provides a typical example. The tyre war involving Firestone, Goodyear and Dunlop during the 1960s had the technological effect of producing wider and taller tyres, better tyre construction, and a greater understanding of compound characteristics. The 'slick' (untreaded) tyre was introduced in 1971 by Goodyear as a means of increasing the contact area of the patch on the road surface. Firestone followed this trend a few months later. Tyre compounds were made softer and, when heated under friction, would grip more efficiently. The width, height and profile of tyres all increased throughout this period as more traction was sought, though eventually changes in regulations brought this trend to a halt.

With the increased understanding of tyre technology and aerodynamics that occurred during the first half of the decade, the sport settled into several seasons of close racing and incremental improvements. In 1977, however, Team Lotus introduced the first of a number of

aerodynamically revolutionary cars that used the Bernoulli principle that accelerating airflow causes a reduction in pressure. This, the principle on which the aerofoil works, was further developed by Lotus in their Type 78 car in which the differences in air pressure acting on the car were used to 'suck' the vehicle onto the ground giving it greater stability and traction [29, p. 208]. These 'ground effect' principles were far from unknown in Grand Prix racing—B.R.M. had attempted to apply them in 1968, March in 1970 and Brabham in 1973—but the technically dominant Lotus achieved a degree of success which obliged competitors, often underfinanced and with imperfect knowledge of the technology, to follow. In 1981, the Fédération Internationale du Sport Automobile (F.I.S.A.) unilaterally banned the 'sliding skirt', an integral part of the ground effect principle, in an attempt to reduce cornering speeds.¹ Apart from causing internal strife within the Grand Prix organizations, the ban forced the constructors to produce cars that mimicked ground effects by other means: e.g. suspension systems and twin chassis that lowered the body onto the ground once the car was mobile. Finally, as a result of efforts to retrieve any workable ground effect, the cars were solidly sprung, a move which put great strain on both driver and machinery.

In addition to the inauguration of the Lotus Type 78, a second important innovation based on well-known technology occurred in 1977, the introduction of a turbocharged engine. Turbocharging technology, incorporated in both production and racing engines, had been used on front-engined Grand Prix cars on many occasions though it was not fully exploited because of the technical problems associated with the high-revving engines involved. In the United States turbocharging had been successfully introduced in 'Indycar' racing in the early 1970s and the resulting engines proved both powerful and reliable. Grand Prix racing differs, however, from that governed by the United States Automobile Club in that it

involves much more acceleration, deceleration and cornering; therefore, the problem of throttle lag (the time the engine requires to respond to the opening of the throttle) imposes disadvantages on turbocharged cars competing in Grands Prix. Turbocharging was, nevertheless, reintroduced by the state-subsidized French Renault Grand Prix team which showed that this technology could provide a practical alternative to the normally aspirated engine. Once the competitive advantage was discovered, it was not long before other teams were developing their own forced induction engines and so began another cycle of competitively-driven technical development [15, pp. 45–50].

In the aftermath of several disastrous accidents, the governing body moved to reduce the speeds of the cars. F.I.S.A. introduced a progressive reduction in fuel tank capacity in the belief that it would force down overall speeds: from 250 litres in 1983, the maximal capacity was reduced to 220 litres in 1984 and 195 litres in 1985. Pit stops were also banned after 1984. These measures forced teams to adopt new strategies of engine management. The sophistication of the turbocharged engine was such that fuel, when available in abundance, was used not only to run the engine but, in addition, by employing the latent heat of a very rich mixture, to cool the turbocharger. The reduction in fuel volume placed greater emphasis on fuel metering and conservation. There was also the added problem of cooling caused by the need to maintain an optimum working temperature in order to maximize the turbocharged engine's performance [30]. Success went to teams such as McLaren TAG which could create an engine system that combined all these new requirements.

In terms of power output, the turbocharged engine produced typically 500 b.h.p. in 1980 rising to 650 b.h.p. in 1983, about 800 b.h.p. in 1984 and up to 900 b.h.p. in 1985. The era of the turbocharged car was at its height, though for financial reasons turbochargers were not

used by all teams. In 1986, the regulations stipulated that only 1.5-litre engines would be permitted, though they might incorporate supercharging. The 3-litre category, which had encouraged so many new teams to enter the formula back in the 1970s, was ended in favour of a highly expensive, technically very sophisticated and fast new type of Grand Prix car typified by the Marlboro McLaren MP4 and the Williams Honda FW09. Between 1986 and 1989 the 'turbo era' was phased out; by 1990, all entrants were required to use normally aspirated engines of up to 3.5 litres cubic capacity.

4. A summing up

The most conspicuous element in the transformation of Grand Prix motor racing since its appearance at the end of the last century is in the design of the cars. The early square shape, open body and thin tyres have given way to a lower, flatter bodied, machine with large, 'slick' tyres. However, this simple comparison belies the deeper changes that have occurred in over 80 years of technical, engineering and, equally important, commercial development. While the spectacle remains the dominant attraction for the paying public, to the providers of the sport, the whole essence of Grand Prix racing has always lain in the development of technology and its practical and economic exploitation.

However, if these considerations have motivated motor sport throughout its history, an important distinction derives from differences in the nature and functions of the once-dominant 'factory teams' and those of the more recent 'constructor' organizations which manage the technical and commercial relationships among the disparate components of the Grand Prix network. From its inception to the mid to late 1950s, the dominant role in Grand Prix motor racing was played by the emergent car manufacturers, part of whose R&D functions were externalized on the race

track, and for whom the sport was principally a means of testing and promoting their own vehicle designs. Since the late 1950s, the diffusion of the entrepreneurial role has brought commercial interests and criteria to the fore. Though individuals have retained the important function of both technological and commercial foresight, as shown by Chapman's part in the establishment of the Lotus/Ford/Cosworth/Player network, entrepreneurial behaviour has been increasingly diffused among each of the participating organizations.

The transfer of technology between motor sport and commercial automobile manufacture has frequently been portrayed as a dynamic process in which radical innovations are developed as state of the art advances which, having won races for their initiators, are introduced into the roadgoing vehicles and marketed to motorists in general. The actual procedures involved are more complex. When manufacturers' R&D functions were located principally at the track, the transfer of technology from racing to roadgoing vehicles was indeed a more straightforward matter. However, most current developments in Grand Prix technology are incremental: the sport produces few genuine breakthroughs compared with its earliest days and technical progress is primarily a matter of the transfer and adaptation of existing methods. This process of continuous innovation is impelled by two sets of strategic concern: that of the sponsoring organizations and that of the constructor organizations which actually participate on the track.

Where the sponsors are members of the automotive or a related industry (Ford, Honda, Elf, Ferodo) their objectives are not only to acquire the core innovative product and process knowledge that accrues from the racing environment but also, sometimes more urgently, the complementary assets which enable them to market their primary products more effectively. Where they are members of quite separate industries (John Player, Benetton) their

objective is solely the acquisition of complementary assets that allow them to promote a corporate marketing image on an international scale through the medium of televised race coverage [31, 32].

The motive of the constructor teams remains that of the 'enthusiasts' who pioneered the new form of motor sport: winning races as a result of technological advance within the framework of regulations governing the sport. The subsequent dissemination of technical knowledge, if it occurs at all, will depend upon the nature and character of the innovation in question, and the strategic advantage of the innovator. The pressure for technical development is dictated by different types of organization from those of Grand Prix's first, manufacturer-dominated phase. A far greater input is nowadays the responsibility of external companies, organizations other than the team entrants and related sporting bodies which have played a decisive role in the commercialization of the sport. These sponsors are essential to the constructors' ability to achieve their own goals and, in return, the teams sell advertising space on their cars to companies wishing to generate international exposure of their corporate name and image.²

Above all, the role of the manufacturer has changed dramatically and has been redefined through the ascendancy of organizations for which no independent role could have been envisaged when motor sport began. While manufacturers' use of the track as a test bed for technical innovations remains important, an involvement in Grand Prix racing as a component of the company's marketing strategy has become the major rationale [35]. The reasons for the change are to be found not only in the strategic marketing imperatives which define and dominate present commercial activity, but in the technological, regulatory and economic influences which have shaped the progression from Grand Prix sport to Grand Prix industry.

Acknowledgements

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Notes

¹ F.I.S.A., known prior to 1970 as the Commission Sportive Automobile (C.S.I.), is the sporting arm of F.I.A., and is responsible for the government of motor sport internationally.

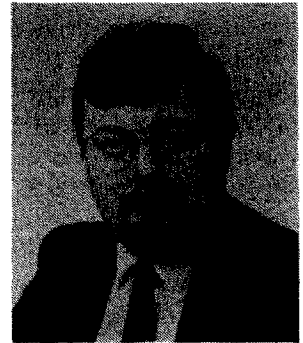
² The Ford/Cosworth/Lotus network was marked by a goal interdependence that ensured a relatively open, stable and co-operative set of relationships. It is not the purpose of the text to suggest that such harmony is inevitable. For details of the less predictable Honda-Williams relationship, see refs 33 and 34.

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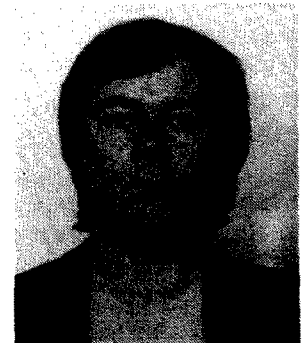
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L'innovation dans les courses Grand Prix d'automobile: l'évolution de la technologie, de l'organisation et de la stratégie

RÉSUMÉ

Depuis son début à la fin du dix-neuvième siècle, les courses automobiles ont été liées à l'industrie de l'automobile. Les courses Grand Prix d'automobiles n'ont pas contribué seulement au transfert de technologie à la fabrication des voitures, mais aussi à la publicité et le développement des automobiles et des pièces détachées. Avant la seconde moitié des années cinquante, ce rapport a été dominée par les fabricants d'automobiles, qu'utilisaient les pistes des courses comme extension de leurs capacités de recherche-développement. Environ 1960, le rapport est devenu plus complexe, caractérisé par un réseau d'organisations des constructeurs et des promoteurs aussi que les fabricants. L'esprit d'entreprise c'est répandu dans ce réseau à la poursuite de plusieurs objectifs: principalement celui de gagner des courses et de transférer des bénéfices promotionnels aux promoteurs. L'utilisation des pistes comme lits d'essai pour l'innovation technique est toujours importante mais devenue secondaire. Des innovations organisationnelles et commerciales aussi que technologiques ont entraîné le développement de l'industrie des automobiles des courses aussi que d'une industrie des Grand Prix.

Innovation bei dem Grand Prix Autorennen: Evolution von Technologie, Organisation und Strategie

ABRISS

Seit dem Beginn des Autorennens am Ende des neunzehnten Jahrhundert besteht eine enge Verbindung mit der Automobilindustrie. Grand Prix Autorennen haben nicht nur beim

Transfer von Technologie zu der Automobilherstellung einen Beitrag geleistet, sondern auch in der Werbung und der Verkaufsförderung von Autos und Bestandteilen eine Rolle gespielt. Bis Mitte der 50iger Jahre war diese Beziehung hauptsächlich von Automobilherstellern bestimmt, die die Rennstrecken als einen Zusatz zu ihren R&D Möglichkeiten betrachteten. 1960 etwa wurde das Verhältnis komplizierter, charakterisiert von einem Netzwerk von Konstrukteuren und Förderern sowie Herstellern. Die Unternehmungsleitung hat sich innerhalb dieses Netzwerkes neue Ziele gesetzt, im besonderen den Sieg im Autorennen und die Übertragung von Werbungsvorteilen auf die Förderer. Der Nutzen der Rennstrecke als Teststrecke für technische Innovation ist weiterhin wichtig aber es ist heutzutage zweitrangig. Organisatorische und kommerzielle wie auch technische Innovation hat zu der Entwicklung des Automobilsports und einer Grand Prix Industrie geführt.

La innovación en las carreras automovilísticas del Grand Prix: la evolución de la tecnología, la organización y la estrategia

RESUMEN

Desde sus comienzos a finales del siglo diecinueve, a las carreras de coches siempre se les ha asociado con la industria automovilística. Las carreras Grand Prix han contribuido no solamente a la transferencia de tecnología hacia la fabricación de los coches particulares, sino también hacia la publicidad y la promoción de coches y de sus componentes. Hasta mediados de los años 50s, dominaban la relación los fabricantes de coches, que empleaban las pistas de carreras como una extensión más de su infraestructura para la investigación y el desarrollo. Hacia el año 1960 la relación existente empezó a cambiar, reflejándose en una red de organizaciones de constructores y auspiciadores a la vez que

fabricantes. Los empresarios se han difundido dentro de este marco, que tiene múltiples objetivos: por encima de todo está el ganar las carreras y hacer beneficiar a los auspiciadores como resultado de la promoción consecuente. El uso de las pistas de carreras como zona de pruebas para la innovación técnica mientras

sigue teniendo todavía su importancia, se queda ya en un segundo plano. Las innovaciones tanto organizacionales como comerciales y también tecnológicas han contribuido al desarrollo por un lado del deporte automovilístico y por el otro de la industria del Grand Prix.